nature chemistry

Article

https://doi.org/10.1038/s41557-024-01505-0

Copper-catalysed asymmetric hydroboration of alkenes with 1,2-benzazaborines to access chiral naphthalene isosteres

Received: 13 May 2023

Accepted: 8 March 2024

Published online: 8 April 2024

Check for updates

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Bioisosteric replacement has emerged as a clear strategy for drug-structure optimization. Naphthalene is the core element of many chiral pharmaceuticals and drug candidates. However, as a promising isostere of naphthalene, the chiral version of 1,2-benzazaborine has rarely been explored due to the lack of efficient synthetic methods. Here we describe a copper-catalysed enantioselective hydroboration of alkenes with 1,2-benzazaborines. The method provides a general platform for the atom-economic and efficient construction of diverse chiral 1,2-benzazaborine compounds (more than 60 examples) that bear a 2-carbon-stereogenic centre or allene skeleton in high yields and excellent enantioselectivities. Three 1,2-benzazaborine analogues of bioactive chiral naphthalene-containing molecules have been prepared, and a series of transformations around chiral 1,2-benzazaborines have also been developed. Notably, the hydroboration process of this study reveals that the identity of 1,2-benzazaborine plays an essential role in the rate-determining step and catalyst resting state.

Bioisosteric replacement is an important strategy for drug discovery that has been widely practised by medicinal chemists[1,2]. Such a strategy can not only improve the physicochemical properties of lead molecules, but can also expand intellectual property space. Naphthalene is a ubiquitous pharmacophore present in many bioactive molecules with anticancer, anti-inflammatory, antimicrobial and antiviral activities, and so on[3]. In recent years, 1,2-benzazaborines have attracted considerable interest as naphthalene replacements[4–8] (Fig. 1a), and often exhibit better bioactivity than their all-carbon equivalents, probably because the NH groups of the 1,2-benzazaborines can act as hydrogen-bond donors for better binding to proteins[9–12] (Fig. 1b). Furthermore, the introduction of three-dimensional elements into flat molecules often acts to improve biological activities and physical properties[13,14]. Naphthalene-containing chiral molecules have a place in marketed pharmaceuticals and drug candidates[3] (Fig. 1c). However, similar molecules with isosteric 1,2-benzazaborines have not been explored because of a lack of efficient methods to assemble such molecules with three-dimensional shape.

Although many direct-synthesis or late-modification methods have been developed to construct substituted 1,2-benzazaborines, the methods for preparing chiral 1,2-benzazaborines are extremely limited[15–18]. The common method for the preparation of 1,2-benzazaborines involves an annulation/aromatization reaction of trifluoroborates and 2-aminostyrenes[19]. However, the methodology is unlikely to provide

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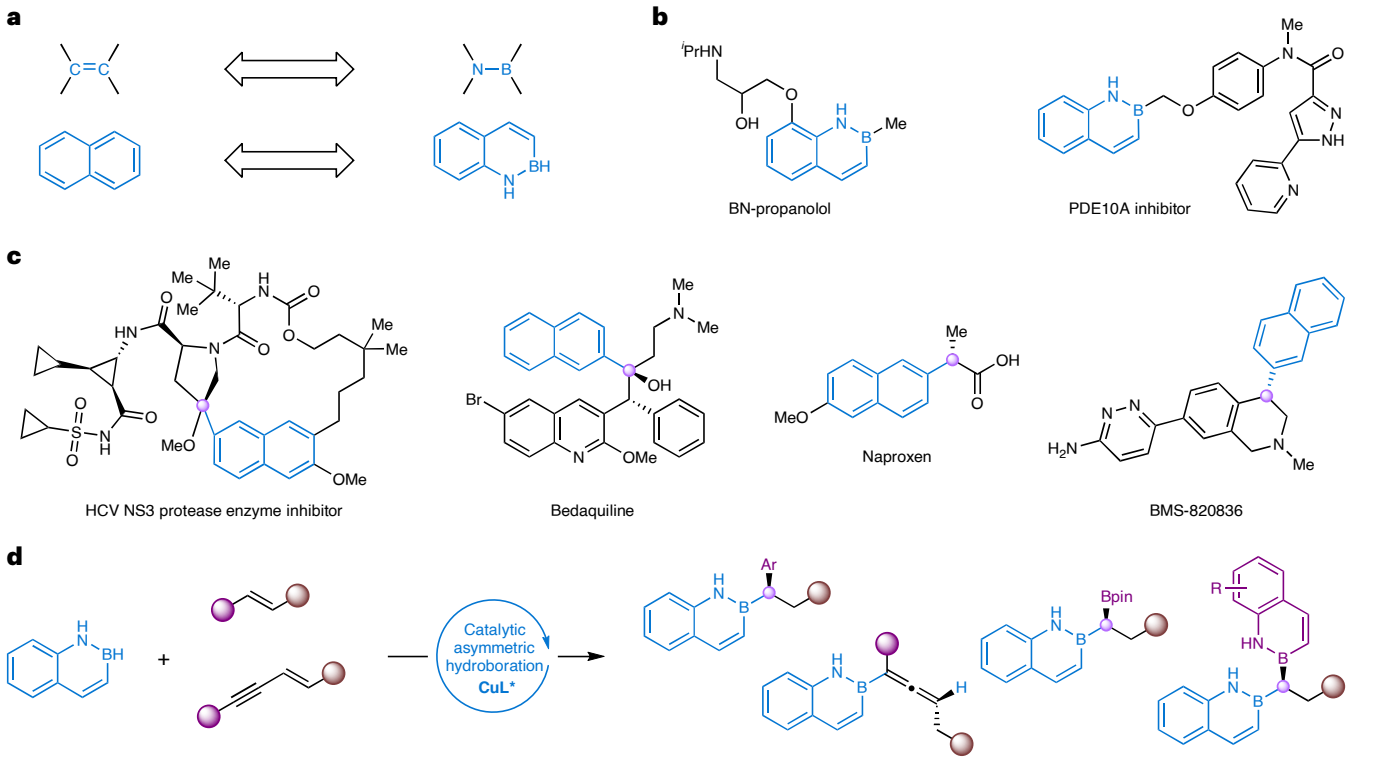


Fig. 1 | Asymmetric hydroboration reaction for the construction of chiral naphthalene isosteres. a, Isoelectronic relationship between naphthalene and 1,2-benzazaborine. **b**, 1,2-Benzazaborine moiety in bioactive molecules. **c**, Commercial drugs or bioactive molecules containing the chiral naphthalene moiety. **d**, This work: copper-catalysed asymmetric hydroboration of alkenes with 1,2-benzazaborines. Bpin, 4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl.

general access to chiral 1,2-benzazaborines due to the lower yields for alkyl trifluoroborates[19], as well as owing to the fact that complex chiral trifluoroborates are difficult to synthesize[20]. A direct asymmetric catalytic strategy for the assembly of such molecules is a much more attractive and promising approach. The catalytic asymmetric hydroboration reaction has emerged as a synthetic toolbox to prepare chiral organoboron compounds[21–41], and we wondered whether such asymmetric hydroboration between alkenes and 1,2-benzazaborines could be used to construct chiral 1,2-benzazaborines. Of note, as an important type of borane, B–H bond activation of 1,2-azaborines involving transition metals has been studied less[18,42], and the corresponding hydroboration reaction has not yet been explored. One main reason might lie in the fact that 1,2-azaborines are more stable than other traditional boranes, such as BH_3, 9-borabicyclo[3.3.1]nonane (9-BBN), pinacolborane (HBpin), catecholborane (HBCat) and 2,3-dihydro-1*H*-naphtho[1,8-*de*][1,3,2]diazaborinine (HBdan)[43]. As evidence, the aforementioned traditional boranes readily react with active protons to release H_2, whereas 1,2-azaborines are relatively stable in acidic environments[18]. Therefore, direct activation of the B–H bond in 1,2-azaborines is a big challenge in boron chemistry, and the successful transformations of this bond will greatly enrich boron chemistry and enhance the utilization of 1,2-azaborines in various fields.

In this Article we describe a simple approach to chiral 1,2-benzazaborines that exploits an asymmetric hydroboration between alkenes and 1,2-benzazaborines (Fig. 1d). This chemistry enables a general and atom-economic preparation of a wide range of chiral 1,2-benzazaborines bearing an α-carbon-stereogenic centre in high yields and excellent enantioselectivities. Moreover, this protocol can also be extended to the efficient preparation of chiral 1,2-benzazaborines with an allene skeleton as well as chiral *gem*-diboryl alkanes. We believe that our strategy will greatly expand the range of chiral organoborons and provide alternative skeletons and efficient methods for medicinal chemistry as well as pharmaceuticals.

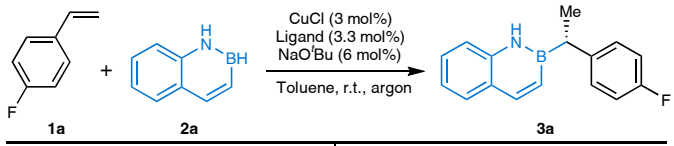
Results and discussion

We first investigated this envisioned asymmetric hydroboration reaction using 4-fluorostyrene (**1a**) and 1,2-benzazaborine (**2a**) as test substrates. As shown in Table 1, a wide range of bisphosphine ligands were investigated for their efficiency to promote the reaction. Pleasingly, the desired product **3a** was obtained in 49% NMR yield and 80% e.e. with CuCl as catalyst and (*R*,*R*)-Me-Duphos (**L1**) as ligand (Table 1, entry 1). This result proved the feasibility of this enantioselective hydroboration using 1,2-benzazaborine as the boron reagent and encouraged us to further investigate other ligands (Table 1, entries 2–8). (*S*,*S*,*S*,*S*)-MeO-BIBOP (**L2**), (*R*,S_p)-Josiphos (**L3**) and (*S*)-QuinoxP (**L4**) were inefficient in this transformation (Table 1, entries 2–4). Axially chiral bisphosphine ligands (*S*)-BINAP (**L5**) and (*R*)-Segphos (**L6**) were almost useless for this reaction (entries 5 and 6). Interestingly, sterically bulky (*R*)-DM-Segphos (**L7**) and (*S*)-DTBM-Segphos (**L8**) were appropriate ligands in terms of both their reaction efficiency and excellent stereocontrol, giving chiral 1,2-benzazaborine **3a** in 65% yield and 93% e.e. (entry 7) and 89% yield and >99% e.e. (entry 8), respectively. **L8** was determined to be the best ligand. Slightly increasing the amount of 1,2-benzazaborine provided optimal conditions for this transformation, affording the desired product **3a** in 94% yield and >99% e.e. (entry 9). A control experiment revealed that the catalytic amount of NaO^{t}Bu was critical for the formation of the desired product (entry 10).

Having optimized the model asymmetric hydroboration of 4-fluorostyrene with 1,2-benzazaborine, we investigated the substrate scope starting with a diverse range of styrenes (Table 2). Variation of the *para*-substituted styrenes (**3a**–**3i**) had little effect on enantioselectivity (96–99% e.e.) for this asymmetric hydroboration reaction, but electron-rich substituents reduced the reactivity (**3e** and **3f**). The absolute configuration of **3b** was determined by X-ray crystallographic analysis (CCDC 2245401). It is worth mentioning that the alkynyl group was tolerated well in moderate yield (**3i**) without additional hydroboration of the alkyne. Various *ortho*- and *meta*-substituted styrenes

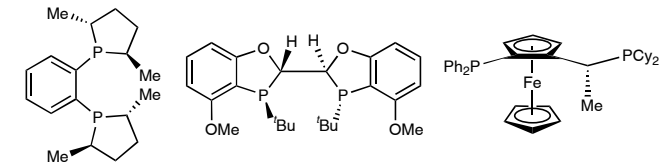
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Table 1 | Condition optimization for the copper-catalysed asymmetric hydroboration reaction

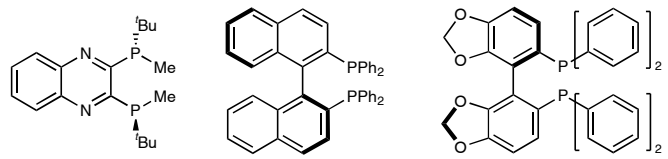


1a **2a** **3a**

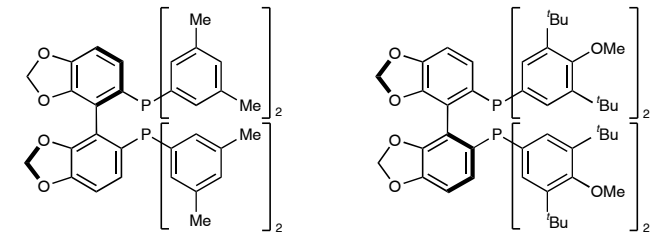
Entry[a]	Ligand	Yield (%)[b]	e.e. (%)[c]	Entry[a]	Ligand	Yield (%)[b]	e.e. (%)[c]
1	**L1**	49	80	6	**L6**	Trace	ND
2	**L2**	11	51	7	**L7**	65	–93
3	**L3**	27	–66	8	**L8**	89	>99
4	**L4**	Trace	ND	9[d]	**L8**	94	>99
5	**L5**	9	44	10[d,e]	**L8**	–	–



(*R*,*R*)-Me-Duphos (**L1**) (*S*,*S*,*S*,*S*)-MeO-BIBOP (**L2**) (*R*,*S*$_p$)-Josiphos (**L3**)



(*S*)-QuinoxP (**L4**) (*S*)-BINAP (**L5**) (*R*)-Segphos (**L6**)



(*R*)-DM-Segphos (**L7**) (*S*)-DTBM-Segphos (**L8**)

[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), CuCl (3 mol%), ligand (3.3 mol%) and NaO^{t}Bu (6 mol%) in toluene at room temperature for 10 h under argon. [b]Yield was determined by ^{1}H NMR analysis. [c]Enantioselectivity was determined by HPLC analysis. [d]**2a** (0.15 mmol). [e]Without NaO^{t}Bu. ND, not determined.

(**3j**–**3m**) were investigated next, and these were also smoothly hydroborated. Strong electron-deficient substituents (**3k**, **3l**) and *ortho*-OMe (**3m**) had a minimal influence on the reactivity and enantioselectivities. There was a low yield of *para*-OMe (**3e**), but *ortho*-OMe (**3m**) may coordinate with the alkyl copper intermediate, resulting in easy insertion of Cu–H species and good yield (35% versus 83%). 2-Vinylnaphthalene was also a suitable candidate for this reaction (**3n**, 99% yield, 98% e.e.). Vinyl-substituted heteroaromatics including benzothiophen (**3o**), benzofuran (**3p**), indole (**3q**) and thiophene (**3r**) all furnished the hydroborated products in moderate to high yields with excellent enantioselectivities (all 98% e.e.). To our delight, diverse β-substituted styrenes (internal alkenes) were also aptly transformed, providing α-selective borated products (**3s**–**3x**) in moderate to good yields with good to excellent enantioselectivities (53–92% yields, up to >99% e.e.). However, β,β-disubstituted styrenes exhibited low reactivity under standard conditions, probably due to difficulty in the addition of Cu–H to the more sterically encumbered olefin. Moreover, this reaction also tolerated more substituted 1,2-benzazaborines (Table 2), affording the enantiopure 1,2-benzazaborine products with excellent enantioselectivities (**3y**–**3aj**, 98% to >99% e.e.). It was found that the electronic property and position of substituents on the 1,2-benzazaborines had little effect on enantioselectivities. Of note, installation of the substituents on the nitrogen atom greatly affected the efficiency of the reaction; the corresponding hydroboration products **3ai** and **3aj** were obtained in 70% yield and 98% e.e. and 37% yield and 98% e.e., respectively, illustrating the steric influence of the substituents of the nitrogen atom on the yield.

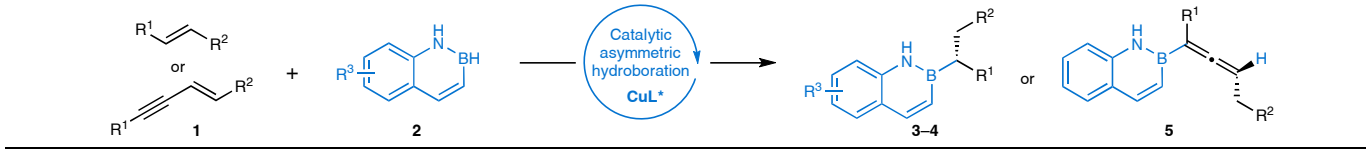
Having established robust reactivity for the asymmetric hydroboration of styrenes, we next turned our attention to explore the enantioselective hydroboration of alkenyl borons (Table 2). A wide range of alkenyl borons could be used as suitable substrates for this addition reaction to deliver the corresponding enantiopure *gem*-diboryl alkanes with excellent enantioselectivities (**4a**–**4i**, 91–98% e.e.). The hydroboration was highly regioselective and only installed 1,2-benzazaborine onto the α-carbon of the boryl group in alkenyl borons. The enantiopure 1,2-benzazaborine containing an α-Bpin group has the potential for further modifications[28,44–47]. Encouraged by the success of the enantioselective hydroboration of alkenyl Bpins, the elusive alkenyl 1,2-benzazaborine substrates were then explored to examine the potential of generating chiral *gem*-di-1,2-benzazaborines. Unlike alkenyl Bpins, the asymmetric addition of alkenyl 1,2-azaborine has not yet been reported. To our surprise, the alkenyl 1,2-benzazaborines could also undergo hydroboration with high efficiency and excellent enantioselectivities to furnish chiral *gem*-di-1,2-benzazaborines **4h** (77% yield, 98% e.e.) and **4i** (87% yield, 98% e.e.).

Having investigated the synthesis of centrally chiral 1,2-benzazaborines, we were interested in exploring the asymmetric hydroboration of 1,3-enynes to access chiral 1,2-benzazaborines with allene skeletons, because allene units are commonly found in natural products, bioactive molecules and materials[48–51]. The enantiopure allenyl 1,2-benzazaborines could be obtained by further simple optimization of the conditions. As shown in Table 2, aryl-, alkenyl- and alkyl-substituted 1,3-enynes could all undergo the asymmetric hydroboration reaction with 1,2-benzazaborines to generate the corresponding chiral allenyl 1,2-benzazaborine products (**5a**–**5g**) in excellent enantioselectivities (97% to >99% e.e.). As can be seen from Table 2, the mild conditions of the asymmetric hydroboration with 1,2-benzazaborines could tolerate diverse functional groups, including halogens (**3a**–**3c**, **3j**, **3t**, **3ab**–**3ae**, **3ai**, **4d**, **4i** and **5c**), ether (**3e**, **3h**, **3m**, **3z**, **3aa**, **4b** and **5a**), ester (**3f**, **3l**, **3aj**, **4c**, **4e** and **5g**), alkyne (**3i**), alkene (**5e**) and imido (**5f**). The absolute configuration of **5b** was determined by computational and experimental circular dichroism spectroscopy (Supplementary Fig. 1).

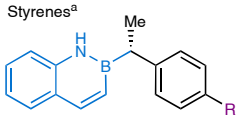
The high functional group tolerance prompted us to explore the utility of late-stage modification of biorelevant molecules (Fig. 2a). A series of bioactive molecules (oestrone, ezetimibe, clofibrate, gemfibrozil, cholesterol and L-menthol) were derivatized into their corresponding alkenes, which, following treatment with 1,2-benzazaborine (**2a**) under established asymmetric hydroboration conditions, were smoothly transformed into eventual chiral 1,2-benzazaborine derivatives **6a**–**6f** with excellent enantioselectivities. The stability of these compounds in air and aqueous conditions was also tested (Supplementary Fig. 3). As isosteres of naphthalene, bioactive molecular analogues **8**, **10** and **12** (refs. 52–54) containing 1,2-benzazaborines could be efficiently synthesized, indicating that this process may be applicable in drug discovery (Fig. 2b). In addition, further synthetic transformations of the chiral 1,2-benzazaborine products were then explored. First, product **3v** could be iodinated to deliver product **13**, and further cross-coupling with organozinc reagents could afford the corresponding 3-arylation product **14** and 3-alkylation product **15** (Fig. 2c). Moreover, intramolecular cyclization of **3t** could take place with lithium *bis*(trimethylsilyl)amide (LiHMDS) to give product **16** in 80% yield without erosion of enantioselectivity (Fig. 2d). Notably, the Bpin group of chiral *gem*-diboryl alkane **4g** could selectively generate tetracoordinated boron species with *n*-BuLi, which could react with tropylium tetrafluoroborate to furnish product **17** (ref. 55), albeit in low

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Table 2 | Substrate scope



Styrenes[a]



3a, R = F, 92% yield, 99% e.e.

3b, R = Br, 70% yield, 98% e.e.

CCDC (2245401)

3c, R = Cl, 84% yield, >99% e.e.

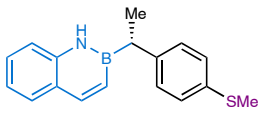
3d, R = Me, 85% yield, >99% e.e.

3e, R = OMe, 37% yield, 98% e.e.

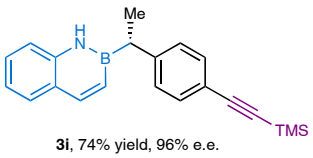
3f, R = OAc, 50% yield, 98% e.e.

3g, R = H, 96% yield, 98% e.e.

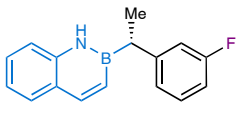
94% yield, 99% e.e. (5 mmol)



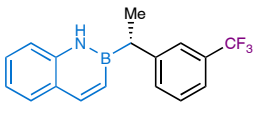
3h, 76% yield, 98% e.e.



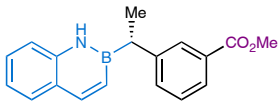
3i, 74% yield, 96% e.e.



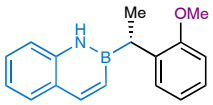
3j, 97% yield, 98% e.e.



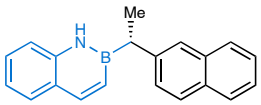
3k, 87% yield, 97% e.e.



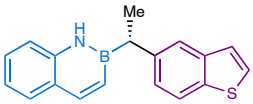
3l, 69% yield, 98% e.e.



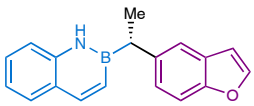
3m, 83% yield, 98% e.e.



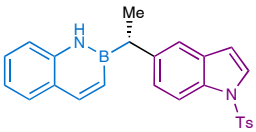
3n, 99% yield, 98% e.e.



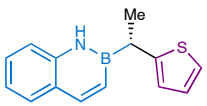
3o, 90% yield, 98% e.e.



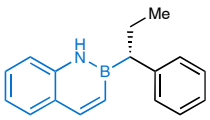
3p, 68% yield, >99% e.e.



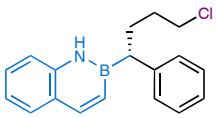
3q, 93% yield, 98% e.e.



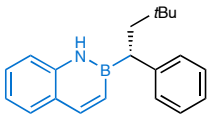
3r, 70% yield, >99% e.e.



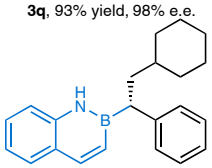
3s, 92% yield, >99% e.e.



3t, 87% yield, >99% e.e.

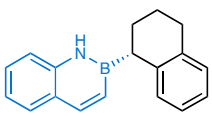


3u, 42% yield, >99% e.e.

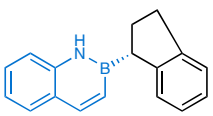


3v, 63% yield, >99% e.e.

55% yield, 99% e.e. (2 mmol)

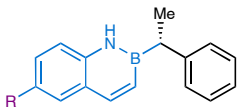


3w, 67% yield, 87% e.e.



3x, 53% yield, 98% e.e.

1,2-Benzazaborines[a]



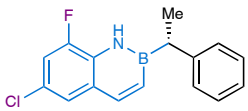
3y, R = Me, 50% yield, 98% e.e.

3z, R = OMe, 67% yield, 98% e.e.

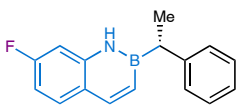
3aa, R = OCF_3, 71% yield, >99% e.e.

3ab, R = Cl, 84% yield, >99% e.e.

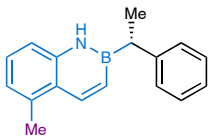
3ac, R = Br, 70% yield, 98% e.e.



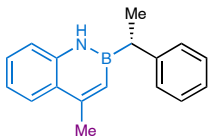
3ad, 49% yield, 98% e.e.



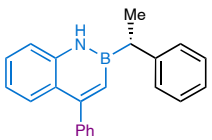
3ae, 92% yield, 98% e.e.



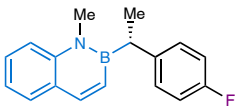
3af, 93% yield, >99% e.e.



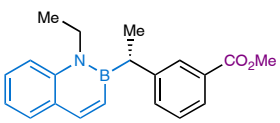
3ag, 96% yield, 99% e.e.



3ah, 77% yield, 98% e.e.

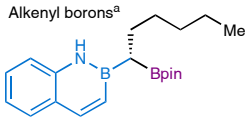


3ai, 70% yield, 98% e.e.

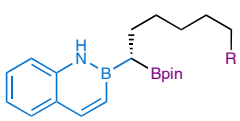


3aj, 37% yield, 98% e.e.

Alkenyl borons[a]

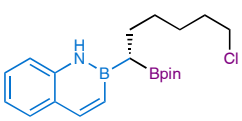


4a, 71% yield, 94% e.e.

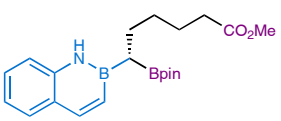


4b, R = OPh, 44% yield, 96% e.e.

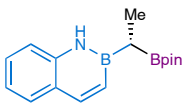
4c, R = OBz, 32% yield, 95% e.e.



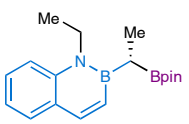
4d, 48% yield, 97% e.e.



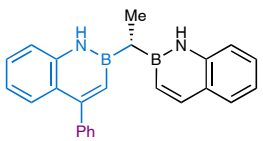
4e, 51% yield, 96% e.e.



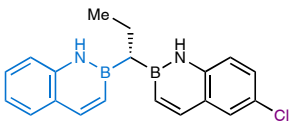
4f, 67% yield, 91% e.e.



4g, 43% yield, 94% e.e.

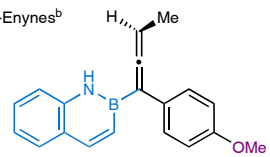


4h, 77% yield, 86% e.e.

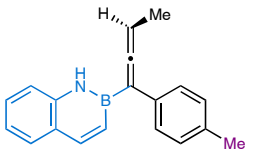


4i, 87% yield, 91% e.e.

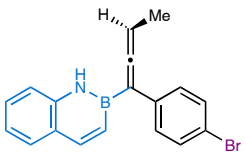
1,3-Enynes[b]



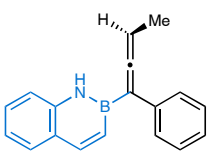
5a, 85% yield, >99% e.e.



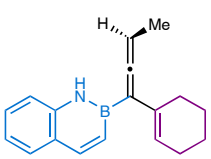
5b, 56% yield, 98% e.e.



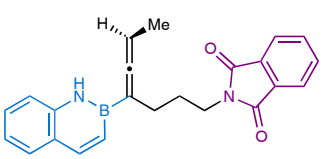
5c, 92% yield, 97% e.e.



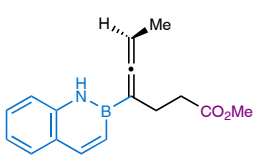
5d, 50% yield, 98% e.e.



5e, 72% yield, >99% e.e.



5f, 54% yield, 97% e.e.



5g, 47% yield, >99% e.e.

[a]Reaction conditions: **1** (0.1 mmol), **2** (0.15 mmol), CuCl (3 mol%), (*S*)-DTBM-Segphos (3.3 mol%) and NaO^{t}Bu (6 mol%) in toluene at room temperature under argon. [b]Reaction conditions: **1** (0.1 mmol), **2** (0.15 mmol), CuOAc (5 mol%), (*S*,*S*)-Ph-BPE (6 mol%) in *n*-hexane/toluene at room temperature under argon. OAc, acetoxy; TMS, trimethylsilyl; Ts, *p*-toluenesulfonyl; OBz, benzoyloxy.

https://doi.org/10.1038/s41557-024-01505-0

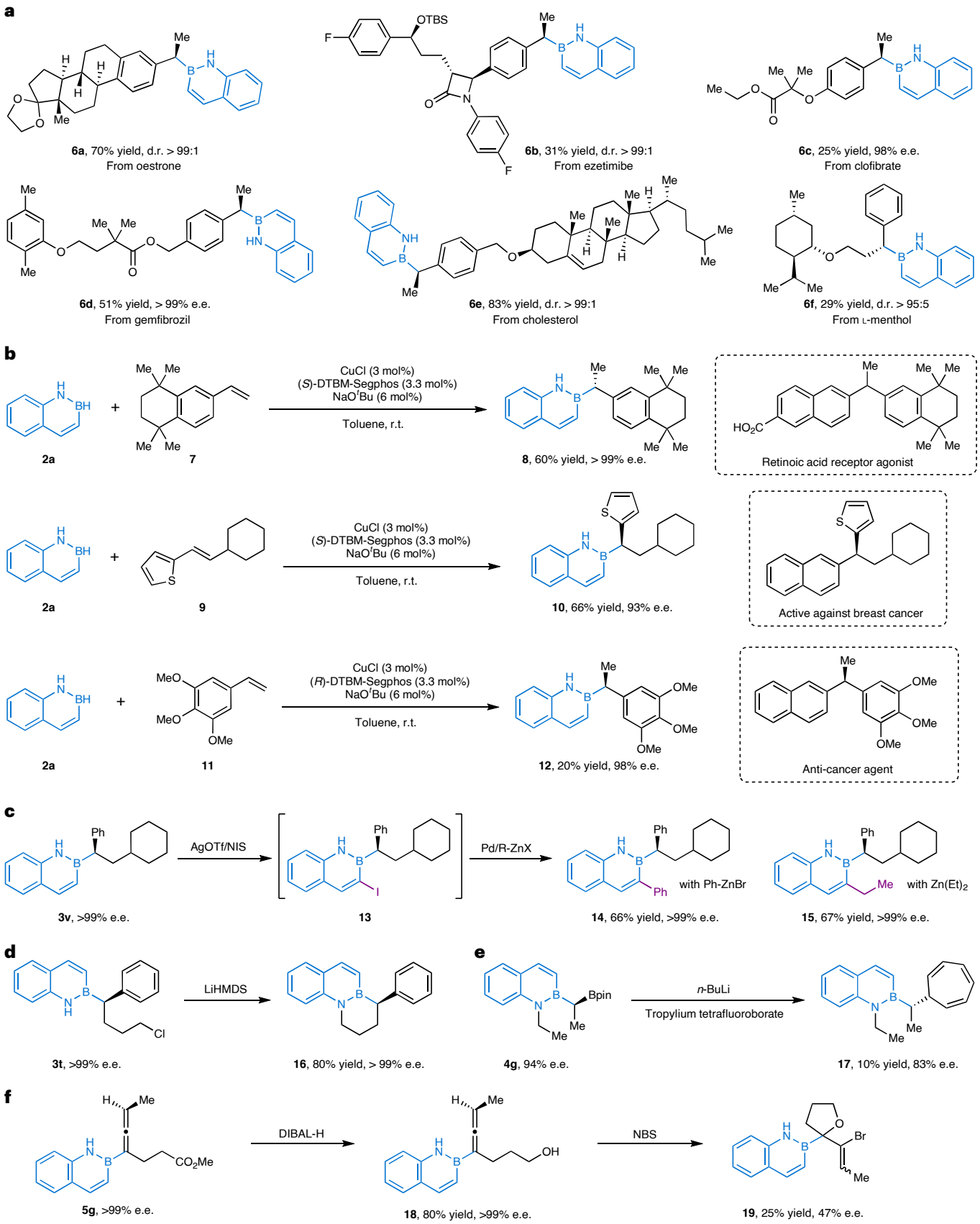
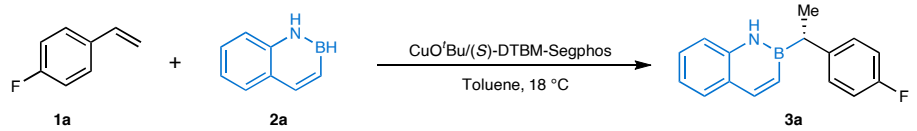


Fig. 2 | Functionalization of complex molecules and synthetic applications. **a**, Late-stage modification of complex molecules. **b**, Synthesis of bioactive molecular analogues. **c**, Ag-catalysed iodization and subsequent Pd-catalysed Negishi cross-coupling of chiral 1,2-benzazaborine **3v**. **d**, Intramolecular cyclization of chiral 1,2-benzazaborine **3t**. **e**, Selective deboronation functionalization through the reaction between tropylium tetrafluoroborate and tetracoordinated boron species from the Bpin group of chiral 1,2-benzazaborine **4g** and *n*-BuLi. **f**, Reduction of the ester group of chiral allenyl 1,2-benzazaborine **5g** and bromoetherification. TBS, *t*-butyldimethylsilyl; NIS, *N*-iodosuccinimide; LiHMDS, lithium *bis*(trimethylsilyl) amide; DIBAL-H, diisobutylaluminum hydride; NBS, *N*-bromosuccinimide.

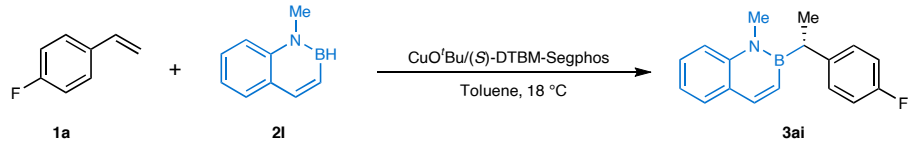
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a



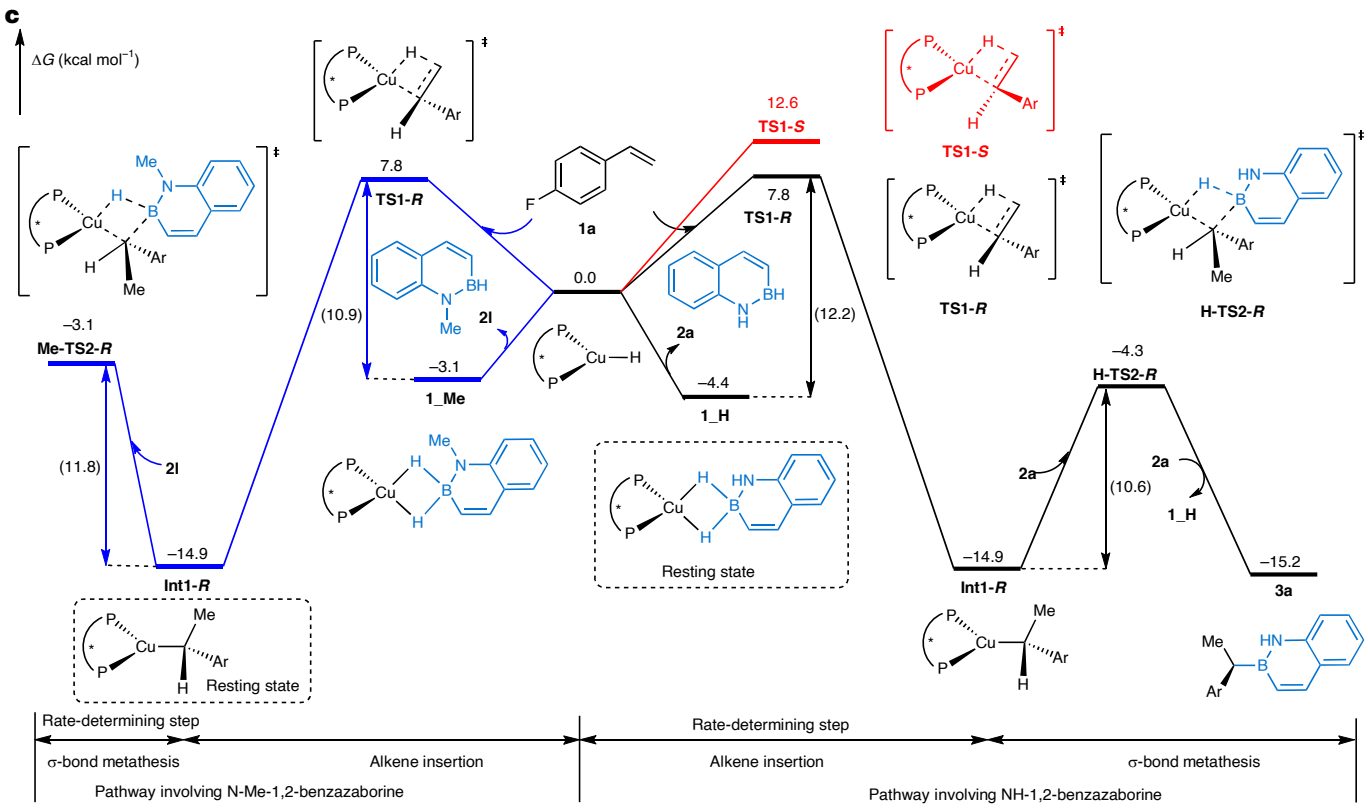
Component	Rate order
1a	First
2a	–0.6
Cu-catalyst	First

b



Component	Rate order
1a	Zero
2l	First
Cu-catalyst	First

c



d

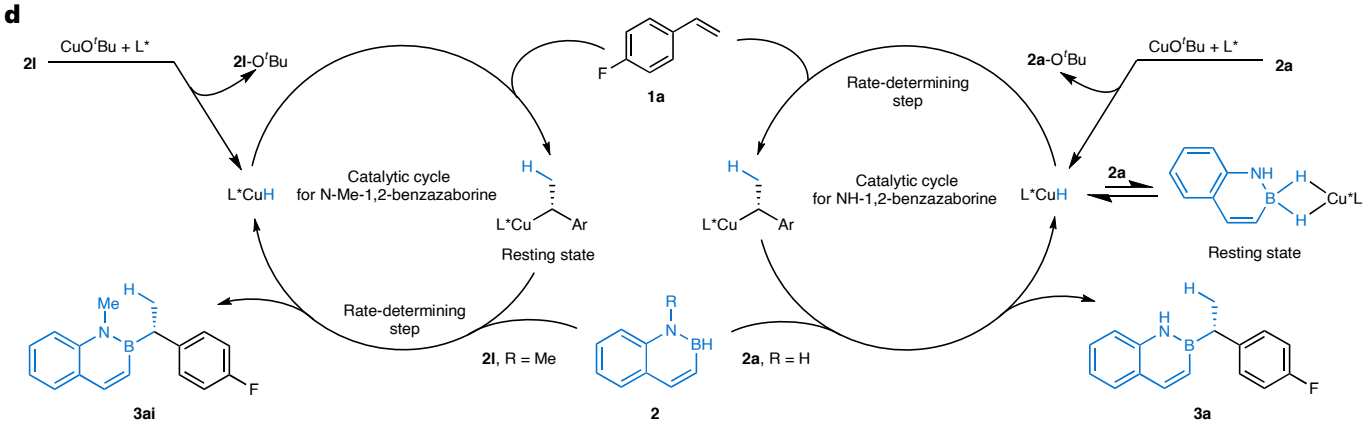


Fig. 3 | Mechanistic investigation. a, Initial rate experiments for alkene with NH-1,2-benzazaborine (**2a**). The experiments indicate that the reaction is first-order in the alkene and Cu-catalyst and negative order in NH-1,2-benzazaborine. **b**, Initial rate experiments for alkene with N-Me-1,2-benzazaborine (**2l**). The experiments indicate that the reaction is zero-order in alkene and first-order in N-Me-1,2-benzazaborine and Cu-catalyst. **c**, DFT calculations: the computed free energy profile for the asymmetric hydroboration of alkene with 1,2-benzazaborine or N-Me-1,2-benzazaborine. Free energies are given in kcal mol^{-1}. **d**, Proposed catalytic cycle for the asymmetric hydroboration of alkenes with 1,2-benzazaborines. Ar, 4-F-C_6H_4.

yield (Fig. 2e). The chiral allene **5g** was readily reduced with DIBAL-H to deliver alcohol **18**, which was followed by a tandem bromoetherification to lead to a tetrahydrofuran-substituted 1,2-benzazaborine **19** (Fig. 2f).

The general copper-catalysed hydroboration process includes alkene insertion into L*Cu–H species, and σ-bond metathesis between the resulting L*Cu–alkyl intermediate and borane to generate the product and regenerate the L*Cu–H species[56]. To gain a deeper insight into the mechanism of this asymmetric hydroboration reaction, a series of experiments were carried out. To achieve better experimental reproducibility, the initial rates of the reaction used CuO^{t}Bu as catalyst. For the hydroboration of NH-1,2-benzazaborine (**2a**), the experimental results showed that the reaction was first-order with respect to alkene **1a** and the Cu-catalyst, indicating that the alkene insertion into L*Cu–H species was probably the rate-limiting step (Fig. 3a; details are provided in Supplementary Figs. 4–6). A negative order (−0.6 order) was found with respect to borane **2a**, which suppressed the alkene insertion process, possibly due to the formation of a resting-state complex between the L*Cu–H species and borane **2a** (Fig. 3a). The generation of such a copper–dihydridoboron complex **1_H** (resting-state complex) was further verified by ^{11}B NMR spectroscopy (details are provided in Supplementary Fig. 10). Interestingly, for the hydroboration of N-Me-1,2-benzazaborine (**2l**), the reaction was zero-order in alkene **1a** and first-order in borane **2l** and the Cu catalyst (Fig. 3b; details are provided in Supplementary Figs. 7–9), indicating that the σ-bond metathesis between the L*Cu–alkyl intermediate and borane **2l** was the rate-determining step. Density functional theory (DFT) calculations were performed to further investigate the mechanism of this reaction (Fig. 3c). For NH-1,2-benzazaborine (**2a**), from the resting state **1_H**, the active L*Cu–H species is formed by releasing a 1,2-benzazaborine molecule in an endergonic process (4.4 kcal mol^{-1}). Subsequently, alkene insertion into the L*Cu–H species occurs via the favourable transition state **TS1-*R*** to generate the L*Cu–alkyl intermediate **Int1-*R***. This process needs to overcome a free energy barrier of 12.2 kcal mol^{-1} relative to **1_H**. Finally, product **3a** is formed by σ-bond metathesis between **Int1-*R*** and **2a** via transition state **H-TS2-*R***, which has a free energy barrier of 10.6 kcal mol^{-1}. Meanwhile, for N-Me-1,2-benzazaborine (**2l**), from the copper–dihydridoboron complex **1_Me** to the transition state **TS1-*R*** of alkene insertion there is a free energy barrier of 10.9 kcal mol^{-1}. Subsequent σ-bond metathesis occurs via transition state **Me-TS2-*R*** with a free energy barrier of 11.8 kcal mol^{-1}, to give final product **3ai**. The calculated results indicate that the rate-determining steps are respectively alkene insertion and σ-bond metathesis for substrates NH-1,2-benzazaborine (**2a**) and N-Me-1,2-benzazaborine (**2l**), consistent with the experimental observations. The switch in the relative magnitude of these two barriers is a combined result due to the reduced stability of the copper–dihydridoboron complex (from −4.4 to −3.1 kcal mol^{-1}) and a slight increase (from 10.6 to 11.8 kcal mol^{-1}) in the σ-bond metathesis for the N-Me-substituted 1,2-benzazaborine substrate **2l**. These results also show that the identity of 1,2-benzazaborine plays a decisive role in the rate-determining step and catalyst resting state. On the basis of the above results and previous reports[56], a catalytic cycle is proposed, as shown in Fig. 3d.

In conclusion, we have developed a copper-catalysed enantioselective hydroboration reaction to access enantioenriched 2-substituted-1,2-benzazaborine compounds from a variety of alkenes and 1,2-benzazaborines. The value of this chemistry was further proven by late-stage modification of complex molecules, the synthesis of bioactive molecular analogues and derivatization of the resulting products. It is expected that this B–H bond activation of 1,2-benzazaborines will be applicable to diverse enantioselective transformations, thereby allowing 1,2-azaborines to serve as progenitors to chiral arene isosteres of broad interest.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41557-024-01505-0.

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Methods

General procedure for the asymmetric hydroboration reaction of alkenes

In a glove box filled with high-purity argon, a 10-ml Young tube was charged with CuCl (0.6 mg, 3 mol%), (*S*)-DTBM-Segphos (7.8 mg, 3.3 mol%), NaO^{t}Bu (1.2 mg, 6 mol%) and 0.8 ml of toluene, then the mixture was stirred for 10 min. After that, a solution of 1,2-benzazaborines **2** (0.3 mmol, 1.5 equiv.) in 0.4 ml of toluene was injected into the Young tube and stirred for 10 min. Finally, a solution of alkenes (0.2 mmol, 1 equiv.) in 0.4 ml of toluene was added. The tube was removed from the glove box and stirred at room temperature (28–36 °C). Upon completion, proper amount of silica gel was added to the reaction mixture. After removal of the solvent, the crude reaction mixture was purified on silica gel (petroleum ether and ethyl acetate) to afford the desired products.

General procedure for the asymmetric hydroboration reaction of 1,3-enynes

In a glove box filled with high-purity argon, a 10-ml Young tube was charged with CuOAc (1.8 mg, 5 mol%), (*S*,*S*)-Ph-BPE (6.0 mg, 6 mol%) and 0.8 ml of hexane, then the mixture was stirred for 10 min. After that, a solution of 1,2-benzazaborines **2** (0.3 mmol, 1.5 equiv.) in 0.2 ml of toluene was injected into the Young tube, and stirred for 10 min. Finally, a solution of 1,3-enynes (0.2 mmol, 1 equiv.) in 0.8 ml of hexane was added. The tube was removed from the glove box and stirred at room temperature (28–36 °C). Upon completion, a proper amount of silica gel was added to the reaction mixture. After removal of the solvent, the crude reaction mixture was purified on silica gel (petroleum ether and ethyl acetate) to afford the desired products.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information files. Crystallographic data for the structure reported in this Article has been deposited at the Cambridge Crystallographic Data Centre, under deposition number CCDC 2245401 (**3b**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures/.

Acknowledgements

Financial support from the National Natural Science Foundation of China (22271048 and 22001038 to K.Y.; 21931013 and 22271105 to Q.S.), the Natural Science Foundation of Fujian Province (2022J05016 to K.Y.; 2022J02009 to Q.S.), Fuzhou University (510578 to Q.S.), Guangdong Provincial Key Laboratory of Catalysis (2020B121201002 to P.Y.) and Shenzhen Science and Technology Program (KQTD20210811090112004 to P.Y.) is gratefully acknowledged. Computational work was supported by the Center for Computational Science and Engineering and the CHEM High-Performance Supercomputer Cluster (CHEM-HPC) of the Department of Chemistry, Southern University of Science and Technology.

Author contributions

Q.S. and K.Y. conceived and directed the project. W.S., J.Z., X.Z. and W.Q. performed experiments. W.S. prepared the Supplementary Information. P.Y. and Y.C. performed the DFT calculations and drafted the DFT parts. Q.S., P.Y. and K.Y. wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41557-024-01505-0.

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Peer review information *Nature Chemistry* thanks Bi-Jie Li, Ying He and the other, anonymous, reviewers for their contribution to the peer review of this work.

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